Complete repair hypergraphs: exact transfer under concatenation

A twisted-cubic–axis family

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Abstract

For a linear code C and a coordinate x, let $\mathcal{H}_x^{(r)}(C)$ be the complete hypergraph of all linear recovery sets of size at most r at x. Its matching number ν_x is disjoint availability, while its transversal number τ_x is the least number of helper failures that block every radius-r repair.

Our principal result is an exact concatenation theorem. If I is an inner code, $r+1 < 2d(I^\perp)$, and the outer functional-dual distance is at least $r+2$, then every bounded repair of the concatenated code lies in one inner block. Thus the entire hypergraph $\mathcal{H}_x^{(r)}$, not merely a selected repair family, is copied block by block. Ordinary extension-field dual distance implies the functional gate through the trace pairing.

For $q = 3^h$, a twisted-cubic–axis code supplies an explicit seed: we compute its coordinatewise matching and transversal numbers and prove $\tau_x > \nu_x$ at every coordinate when $q \geq 9$. Combining the field-nine seed with a self-dual Tsfasman–Vlăduţ–Zink outer family gives unbounded $\mathbb{F}_9$-linear codes of positive rate and relative distance with the exact radius-three repair profiles preserved in every block. Restoring the cubic point at infinity gives a more symmetric radius-four comparison family.

1 Introduction

Locality asks how many surviving symbols are sufficient to recover one lost coordinate of a code. Availability refines locality by asking for many pairwise-disjoint recovery sets. These parameters have become standard in the theory of locally repairable codes; see, for example, Gopalan–Huang–Simitci–Yekhanin [4] and the geometric constructions of Pamies-Juarez–Hollmann–Oggier [11] and Lavrauw–Van de Voorde [9].

Disjoint availability does not determine robustness against adversarial helper failures. The full family of recovery sets may overlap in a way that admits a small blocking set, or it may remain repairable after every set of failures much larger than a maximum disjoint family. The natural object is therefore the complete bounded repair hypergraph at a coordinate. We write ν for its matching number and τ for its transversal number. Then ν is disjoint availability, while $\tau - 1$ is the maximum number of arbitrary helper failures that can always be tolerated while retaining at least one local repair. Our emphasis is the exact coordinatewise computation of both invariants for a complete code-derived repair family and their simultaneous transfer.

Our inner families come from a classical object in finite projective geometry. In $\mathrm{PG}(3, q)$, take the affine part of a twisted cubic and, when q has characteristic three, adjoin the common axis of the osculating planes. We study both this punctured system and the system obtained by restoring the projective cubic point at infinity. The choice is driven by its small dependencies: triples on the axis give two-helper repairs, while pairs of cubic points complete with an axis point to give three-helper repairs. The axis and its stabilizer orbits are classical; a modern account appears in Davydov–Marcugini–Pambianco [2]. What matters here is not an orbit classification but the complete system of small linear dependencies through each selected point.

The second ingredient is concatenation. Concatenation with a locally repairable inner code is established machinery and yields asymptotically good LRC families; Liu–Ma–Wu–Xing [10], in particular, derive locality and asymptotic bounds this way. Locality preservation only asserts that some inner repairs survive. We isolate a stronger dual-distance condition that prevents new low-weight dual words from spanning several inner blocks. Under this condition, *every* bounded repair support in the concatenated code is an embedded inner repair support. Thus matching, transversal, edge counts, and all other invariants of the complete repair hypergraph transfer exactly.

The theorem hierarchy is deliberate. The principal general result is the complete-hypergraph transfer theorem, Theorem 6.1. Theorems 4.1 and 4.2 supply the explicit seed on which $\tau > \nu$ holds at every coordinate, and Theorem 8.1 is their fixed-alphabet synthesis. The projectively completed seed is a more symmetric comparison that transfers one additional repair radius at a modest rate cost.

Contributions

result	mechanism	payoff
Explicit seed	Complete small-circuit classification; cap-set reduction for axis coordinates and a rainbow matching for cubic coordinates	Exact all-symbol rows with $\tau_x > \nu_x$ for $q \geq 9$; three concrete field-nine profiles
Exact transfer	Inner block confinement plus an outer functional-dual distance gate; sharpness examples over $\mathbb{F}_3$	Equality of every complete bounded repair hypergraph, hence preservation of all its invariants and every exact locality below the radius
Fixed-alphabet synthesis	Trace duality and a self-dual TVZ outer family	Unbounded $\mathbb{F}_9$ families with positive rate and distance and exact blockwise repair profiles; a radius-four completed comparison

Table 1: The theorem hierarchy: explicit seed, reusable transfer theorem, and asymptotic synthesis.

Sections 2–5 develop the first row of Table 1. Sections 6–8 prove the transfer mechanism, connect its functional hypothesis to ordinary dual distance, and derive the headline and comparison families.

Scope of the contribution

The underlying geometry, repair tolerance as a hitting number, and locality-preserving concatenation are prior art. The contributions studied here are narrower: exact all-symbol matching and transversal numbers for this point system, and equality of its *complete* bounded repair hypergraph under an outer-dual gate. Section 9 gives the detailed comparison with adjacent work.

2 Complete bounded repair hypergraphs

Let $C \le \mathbb{F}_q^X$ be a linear code on a finite coordinate set X, with dual $C^\perp$ under the standard dot product.

Definition 2.1. Fix $x \in X$ and $r \ge 0$. The *complete radius-r repair hypergraph* at x records exactly the helper support of every dual relation involving x and at most r other coordinates:

$$\mathcal{H}_x^{(r)}(C) = \left\{ R \subseteq X \setminus \{x\} : |R| \le r, \begin{array}{l} \text{there is } y \in C^\perp \text{ with } y_x \ne 0, \\ \operatorname{supp}(y) = \{x\} \cup R \end{array} \right\}.$$

Its inclusion-minimal members form the repair clutter $\mathcal{H}_{x,\min}^{(r)}(C)$. We put

$$\nu_x^{(r)}(C) = \nu(\mathcal{H}_x^{(r)}(C)), \qquad \tau_x^{(r)}(C) = \tau(\mathcal{H}_x^{(r)}(C)),$$

where ν is maximum matching size and τ is minimum transversal size.

The word y supplies the recovery equation

$$c_x = -y_x^{-1} \sum_{z \in R} y_z c_z \qquad (c \in C).$$

Thus Definition 2.1 includes every linear repair whose coefficient support has at most r helpers; it is not a declared or selected subfamily.

Proposition 2.2. *For every finite code-derived repair hypergraph with nonempty edges,*

$$\nu_x^{(r)}(C) \le \tau_x^{(r)}(C).$$

Passing to the inclusion-minimal repair clutter preserves both ν and τ. The dual distance also controls edge uniformity: if $r + 1 \le d(C^\perp)$, every edge has exactly r helpers.

Proof. A transversal meets every edge of a matching, and the matching edges are disjoint, proving $\nu \le \tau$. Every edge contains an inclusion-minimal edge. Consequently a set hits all edges exactly when it hits all minimal edges; replacing each matching edge by a minimal subedge preserves disjointness. Finally, an edge R is witnessed by a nonzero dual word of weight $|R| + 1$. Dual distance gives $r + 1 \le |R| + 1$, while the definition gives $|R| \le r$. $\square$

The operational meaning of τ is immediate: every failure set of size at most $\tau - 1$ misses some repair edge, whereas a transversal of size τ blocks them all. Matching and transversal can nevertheless be far apart; equality is not a generic hypergraph law. Figure 1 illustrates the distinction on the smallest graph example.

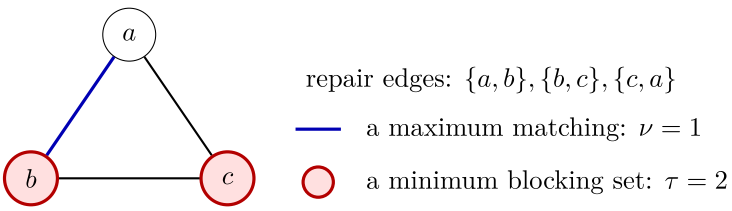

Figure 1: Matching and transversal measure different failure modes. In this three-repair example no two repairs are disjoint, but two helper failures are needed to block them all.

Remark 2.3 (Operational scope). The displayed equation gives one direct scalar protocol: each helper forwards its stored $\mathbb{F}$-symbol, so access and download are both $|R|$ field symbols. This is achieved scalar help-by-transfer, not a bandwidth or access optimum in a subpacketized model [12, 6, 15]. Nor are the raw coefficients invariants: diagonal column rescaling changes their ratios while preserving every dual support. The results below transfer support invariants such as disjoint availability and failure tolerance, not optimal bandwidth, optimal access, or arithmetic cost.

3 The characteristic-three twisted-cubic–axis code

We seek a projective system with two complementary kinds of small dependencies: collinear triples for locality two, and mixed cubic–axis circuits for locality three. Characteristic three supplies both in one classical configuration.

Let $q = 3^h$ and write $\mathbb{F} = \mathbb{F}_q$. In homogeneous coordinates on $\mathrm{PG}(3, q)$ define

$$T_q = \{C(t) = (1 : t : t^2 : t^3) : t \in \mathbb{F}\}$$

and

$$L_q = \{A(u) = (0 : 1 : u : 0) : u \in \mathbb{F}\} \cup \{A_\infty = (0 : 0 : 1 : 0)\}.$$

The set T_q is the affine twisted cubic and L_q is the full line $X_0 = X_3 = 0$. For $q \equiv 0 \pmod 3$, this is the common axis of the osculating-plane pencil [2, Section 7]. We use that geometric name only as provenance: all proofs below start from the displayed coordinates.

Let $S_q = T_q \sqcup L_q$ and let G_q be the $4 \times (2q+1)$ matrix whose columns are fixed representatives of these points. Write C_q for the row code of G_q. Changing a projective representative only rescales one code coordinate, so the repair supports are unaffected.

Theorem 3.1 (Code parameters)**.** *For every finite field of characteristic three,*

$$C_q = [2q + 1, 4, q - 1]_q.$$

Proof. The length is $2q+1$, and, for example, $C(0), C(1), A(0), A_\infty$ span $\mathbb{F}^4$. The distance of a projective-system code is its length minus the largest hyperplane section. If a plane does not contain L_q, it meets L_q in at most one point and its equation restricts to a nonzero cubic polynomial on T_q, so it contains at most three points of T_q. If it contains L_q, its equation has the form $aX_0 + dX_3 = 0$. On T_q this becomes $a + dt^3 = 0$, which has at most one solution because Frobenius $t \mapsto t^3$ is injective. Hence every plane section has size at most $q + 2$. The plane $X_3 = 0$ contains all of L_q and the point $C(0)$, attaining $q + 2$. Therefore $d = (2q + 1) - (q + 2) = q - 1$. □

3.1 Small circuits and exact locality

For distinct $s, t, u \in \mathbb{F}$, put

$$e_1 = s + t + u, \qquad e_2 = st + su + tu.$$

These symmetric sums are precisely the coordinates of the axis point that completes three cubic points to a dependency.

Theorem 3.2 (Small-circuit classification)**.** *The inclusion-minimal dependent subsets of S_q having size at most four are exactly:*

(a) the triples of distinct points of L_q;

(b) the sets

$$\{C(s), C(t), C(u), (0 : e_1 : e_2 : 0)\}$$

for distinct $s, t, u \in \mathbb{F}$.

The pair (e_1, e_2) is never $(0, 0)$ for distinct s, t, u, so the axis completion in (b) is a unique projective point. Consequently every axis coordinate has exact locality two and every cubic coordinate has exact locality three.

Proof. Every three distinct points on L_q are minimally dependent. Vandermonde independence excludes all-cubic sets of size at most four because the cubic point at infinity has been punctured. A set with one cubic point and at most two axis points is independent by its X_0 coordinate and the independence of two points on L_q; adding a third axis point makes it nonminimal. A set with two cubic points and at most one axis point is independent from the X_0, X_3 coordinates and Frobenius injectivity. With two axis points $A(v), A(w)$, its determinant is, up to sign,

$$(t-s)^3(w-v),$$

which is nonzero for distinct parameters. These cases exhaust all mixed sets except three cubic points and one axis point.

Write that axis point as $(0 : \alpha : \beta : 0)$. Expanding along its column gives, up to the nonzero Vandermonde factor $(s-t)(s-u)(t-u)$, the scalar

$$\alpha e_2 - \beta e_1.$$

Thus dependence holds exactly at $(\alpha : \beta) = (e_1 : e_2)$, establishing both existence and uniqueness. If $e_1 = e_2 = 0$, then s, t, u would be the three roots of $X^3 - stu$; injectivity of Frobenius would force them equal. Thus the four points form a circuit. The locality assertions follow from the dual-support/circuit correspondence and the absence of smaller circuits through the respective target type. □

4 Matching and transversal invariants

Let $Z_3(q)$ denote the largest size of a subset of the additive group of $\mathbb{F}_q$ containing no three distinct elements with sum zero. Under an $\mathbb{F}_3$-basis, this is the usual affine cap number in $\mathrm{AG}(h, 3)$. It enters because complements of axis-repair transversals are precisely such zero-sum-free sets.

4.1 Axis coordinates

Fix $y \in L_q$. The minimal radius-three repair clutter at y splits over two disjoint helper grounds, one containing only axis helpers and the other only cubic helpers:

- every pair among the q other axis points repairs y, giving the complete graph K_q;
- a cubic triple repairs y exactly when its completion in Theorem 3.2 equals y.

Matching and transversal numbers add across this disjoint union.

For A_∞, the cubic triples are exactly the distinct zero-sum triples of $\mathbb{F}_q$, hence the affine lines of $\mathrm{AG}(h, 3)$. For $A(a)$, the following projective relabeling moves the target to A_∞:

$$t \longmapsto (t+a)^{-1}, \qquad 0^{-1} = 0.$$

This convention at the pole $t = -a$ makes the map a permutation of $\mathbb{F}_q$. Under it, the cubic repair component becomes the zero-sum triples avoiding zero. Indeed, a triple s, t, u completes to $A(a)$ exactly when $e_2 = ae_1$. None of s, t, u can equal $-a$: substituting one such value would force another to equal it. Hence

$$(s+a)^{-1} + (t+a)^{-1} + (u+a)^{-1} = \frac{e_2 - ae_1}{(s+a)(t+a)(u+a)} = 0.$$

The converse follows by reversing the same calculation.

Theorem 4.1 (Exact axis rows)**.** *At the point A_∞,*

$$\nu_{A_\infty} = \frac{5q-3}{6}, \qquad \tau_{A_\infty} = 2q-1-Z_3(q).$$

At every finite axis point $A(a)$,

$$\nu_{A(a)} = \frac{5q-9}{6}, \qquad \tau_{A(a)} = 2q-2-Z_3(q).$$

These are the invariants of both the minimal repair clutter and the complete radius-three repair hypergraph.

Proof. The complete graph K_q contributes matching $(q-1)/2$ and transversal $q-1$. The affine-line hypergraph contributes matching $q/3$ and transversal $q-Z_3(q)$: parallel lines give the matching, and complements exchange transversals with zero-sum-free sets, since a complement contains no repair triple exactly when the removed set hits every one. After forbidding zero, one parallel class loses its line through zero. Translation preserves the zero-sum condition in characteristic three, and a translate of a maximum zero-sum-free set can be chosen to avoid zero; hence its maximum size remains $Z_3(q)$. The values become $q/3-1$ and $q-1-Z_3(q)$. Adding the contributions gives the formulas. Proposition 2.2 passes the invariants between the complete hypergraph and its minimal clutter. $\square$

The strict gap does not require a quantitative cap-set bound. Any matching in the minimal axis clutter consumes two axis helpers for each pair edge and three cubic helpers for each cubic edge. Weighting axis helpers by three and cubic helpers by two assigns total weight six to either edge type, and yields $6\nu_y \le 5q$. The K_q component alone forces $\tau_y \ge q-1$. Thus $\nu_y < \tau_y$ for $q \ge 9$.

4.2 Cubic coordinates

Fix $x \in \mathbb{F}$. By Theorem 3.2, every unordered pair $\{s,t\} \subseteq \mathbb{F} \setminus \{x\}$ gives one repair

$$\{C(s), C(t), A(\kappa_x(s,t))\},$$

where $\kappa_x(s,t)$ is the unique completion color. For fixed s, the map $t \mapsto \kappa_x(s,t)$ is injective away from the diagonal. The repair hypergraph is therefore a properly edge-colored K_{q-1} in which every graph edge is augmented by its color vertex on the axis.

Theorem 4.2 (Exact cubic rows)**.** *Every cubic coordinate has exactly $\binom{q-1}{2}$ radius-three repairs and*

$$\nu_x = \frac{q-1}{2}, \qquad \tau_x = q-2.$$

In particular, $\tau_x > \nu_x$ for $q \ge 9$.

Proof. The edge count is the number of unordered pairs of cubic helpers. A matching uses two distinct cubic helpers per edge, giving $\nu_x \le (q-1)/2$. For the reverse inequality, we relabel the cubic helpers so completion colors are injectively determined by sums, then choose a perfect matching with distinct sums. Put

$$u = (s-x)^{-1}, \qquad v = (t-x)^{-1}.$$

As s,t range over the cubic helpers, u,v range over $\mathbb{F}_q^*$. Direct simplification in characteristic three shows that the completing axis point is

$$\begin{cases} A_\infty, & u+v=0, \\ A(-x+(u+v)^{-1}), & u+v \ne 0. \end{cases}$$

This is an injective recoding of the sum $u+v$. Choose a generator g of $\mathbb{F}_q^*$ and pair g^{2i} with g^{2i+1} for $0 \le i < (q-1)/2$. These pairs partition $\mathbb{F}_q^*$, while their sums $(1+g)g^{2i}$ are distinct. If $q > 3$, then $g \neq -1$ because g has order $q-1 > 2$, so these sums are also nonzero. For $q = 3$ there is only one pair, so the conclusion is immediate even though its sum is zero. Mapping back by $s = x + u^{-1}$ gives a rainbow perfect matching of the $q-1$ cubic helpers and therefore $\nu_x = (q-1)/2$.

For the transversal upper bound, choose one cubic helper $a \neq x$ and take all cubic helpers other than x and a; this set has size $q-2$ and every repair uses one of them. Conversely, suppose a transversal leaves a nonempty set U of cubic helpers unchosen. Fix $a \in U$. For each $b \in U \setminus \{a\}$, the edge on $\{a, b\}$ forces its completion color into the transversal, and properness makes these colors distinct. Counting the chosen cubic helpers together with these forced colors gives at least $q-2$ vertices. ☐

Combining the two coordinate types gives the uniform statement.

Corollary 4.3 (All-symbol separation)**.** *For every $q = 3^h \ge 9$ and every coordinate z of C_q,*

$$\nu_z^{(3)}(C_q) < \tau_z^{(3)}(C_q).$$

Moreover, C_q has all-symbol locality three, with exact locality two on L_q and exact locality three on T_q.

4.3 The exact field-nine seed

For $q = 9$, the affine cap number is $Z_3(9) = 4$, and the general cubic formula gives $\nu_x = 4$. We obtain the complete table below; repair counts refer to inclusion-minimal repairs, while (ν, τ) is unchanged for the complete hypergraph.

coordinate type	multiplicity	locality	minimal repairs	ν	τ
cubic $C(x)$	9	3	28	4	7
finite axis $A(a)$	9	2	$36+8$	6	12
axis A_∞	1	2	$36+12$	7	13

Table 2: Exact coordinate profiles of the $[19, 4, 8]_9$ seed. Repair counts are inclusion-minimal; $36+8$ and $36+12$ separate the axis-pair and cubic components.

The 36 pair repairs are the edges of K_9. The cubic components contain eight zero-sum triples avoiding zero at a finite axis point and twelve zero-sum triples at A_∞. Globally, the small-circuit inventory consists of $\binom{10}{3} = 120$ axis triples and $\binom{9}{3} = 84$ completed cubic four-circuits.

These rows also give the uniform ratio

$$7\nu_z \le 4\tau_z.$$

The constant $7/4$ is attained at every cubic coordinate.

5 The projectively completed seed

Restoring the missing cubic point has two effects: it recovers projective symmetry within each parameter line, and it makes the complete inclusion-minimal repair clutter accessible through the ambient rank-four bound.

Let
$$\overline{T}_q = T_q \cup \{C_\infty = (0:0:0:1)\}, \qquad \overline{S}_q = \overline{T}_q \sqcup L_q,$$
and let $\overline{C}_q$ be the row code of the displayed projective system. Thus $\overline{C}_q$ restores the one cubic point punctured from C_q; the axis is unchanged.

Theorem 5.1 (Completed code and symmetry). *For every $q = 3^h$,*
$$\overline{C}_q = [2q+2, 4, q]_q, \qquad d(\overline{C}_q^\perp) = 3.$$
Cubic coordinates have exact locality three and axis coordinates exact locality two. An explicit monomial automorphism is transitive separately on the completed cubic $\overline{T}_q$ and on the axis L_q.

Proof. A plane not containing the axis meets it once and meets the full twisted cubic in at most three points. A plane containing the axis has equation $aX_0 + dX_3 = 0$. If $d = 0$, it contains C_∞ and no finite cubic point; if $d \neq 0$, the equation $a + dt^3 = 0$ has one finite solution and C_∞ is excluded. Hence every section has size at most $q+2$, and the old extremal section still attains $q + 2$. This gives distance $(2q + 2) - (q + 2) = q$; the same spanning columns give dimension four. Pairwise projective independence gives dual distance at least three, while any three distinct axis points give equality.

The new boundary four-circuit
$$\{C(s), C(t), C_\infty, A(s+t)\} \qquad (s \neq t)$$
supplies the cubic-infinity repairs. The shifted-inversion relabeling used above extends projectively across its pole through the invertible map
$$(X_0, X_1, X_2, X_3) \longmapsto (a^3X_0 + X_3, a^2X_0 - aX_1 + X_2, aX_0 + X_1, X_0).$$
Away from $s = -a$, it sends $C(s)$ to $C((s+a)^{-1})$ up to the nonzero scale $(s+a)^3$; it sends $C(-a)$ to C_∞ and C_∞ to $C(0)$. On the axis it sends $A(y)$ to $A((y-a)^{-1})$ up to scale, with $A(a)$ sent to A_∞. Consequently the ambient map induces a monomial code automorphism that is transitive separately on the completed cubic and on the axis. Locality and the repair hypergraph therefore need be calculated at only the two infinity targets. ☐

Theorem 5.2 (Full minimal repair clutter). *Every inclusion-minimal repair in $\overline{C}_q$ uses at most four helpers, so the full minimal repair clutter is already visible at radius four. Its exact rows are*
$$(\nu_x^{(4)}, \tau_x^{(4)}) = \begin{cases} ((q-1)/2, q-1), & x \in \overline{T}_q, \\ ((5q-3)/6, 2q-3), & x \in L_q. \end{cases}$$
These values are uniform within each of the two projective parameter lines.

Proof. For exhaustiveness, take an inclusion-minimal target-plus-helper relation. Its helper columns are linearly independent: otherwise a helper-only relation can be subtracted to cancel one helper without cancelling the target coefficient, because that helper-only relation has target coefficient zero. Since the generator has four rows, there are at most four helpers. This proves that radius four is the full *minimal* inner repair clutter; it makes no assertion about all minimal repairs of a later concatenated code.

For a cubic target, every minimal edge consumes at least two of the q other cubic points, so $\nu \leq (q-1)/2$; the radius-three rainbow matching attains equality. For an axis target, if an edge uses c cubic and a axis helpers, the short-circuit exclusions imply: if $a = 0$ then $c \geq 3$; if $a = 1$ then $c \geq 2$; and if $a \geq 2$ the inequality is immediate. Hence $2c + 3a \geq 6$. Summing over a matching and using the $q+1$ cubic and q available axis resources gives $6\nu \le 5q+2$, hence $\nu \le (5q-3)/6$; the radius-three matching attains equality.

Finally, consider a transversal of the full minimal repair clutter. Its complement contains no helper set that spans the target: any spanning set contains an inclusion-minimal spanning subset, and the preceding rank argument gives that subset at most four helpers. The complement therefore lies in a hyperplane avoiding the target; conversely every such hyperplane complement is a transversal. Equivalently,

$$\tau_x = (2q+1) - m_x,$$

where m_x is the largest hyperplane section avoiding x. The maximum is $q+2$ for a cubic target: after moving the target to C_∞, the global section bound established above applies, and the plane $X_3 = 0$ avoids C_∞ while containing all $q+1$ axis points and $C(0)$. For an axis target, move it to A_∞. Any avoiding plane has nonzero X_2 coefficient, so it contains at most one axis point and at most three cubic points. The plane $X_2 - X_1 = 0$ attains four: it contains $C(0), C(1), C_\infty$, and $A(1)$, but not A_∞. Thus $m_x = 4$. The two transversal values are therefore $q-1$ and $2q-3$. □

At $q = 9$, the completed seed is $[20, 4, 9]_9$. Its ten cubic coordinates all have radius-four row $(4, 8)$, while its ten axis coordinates all have row $(7, 15)$. The two old affine axis subtypes merge because the restored projective symmetry acts transitively on the full axis.

6 Exact repair transfer under concatenation

Informally, each outer symbol is a vector encoded as one inner block. The question is whether a small dual relation can couple several such blocks. We formulate the answer without choosing coordinates on the outer symbol space. Let $I \le \mathbb{F}^K$ be an inner code and let V be an $\mathbb{F}$-vector space equipped with a linear isomorphism $e : V \to I$. For an outer $\mathbb{F}$-linear code $O \le V^J$, the concatenated code $O \circ_e I \le \mathbb{F}^{J\times K}$ encodes each outer symbol through e. Figure 2 previews the separate roles of the two transfer gates.

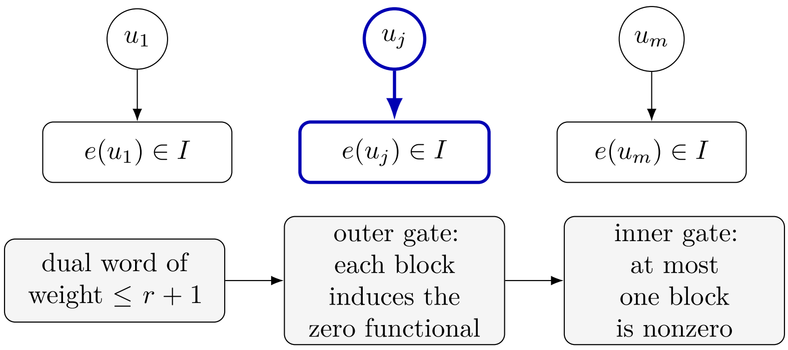

Figure 2: Concatenation expands each outer symbol into an inner block. The outer gate removes cross-block functional coupling; the inner gate then confines every bounded dual relation, and hence every bounded repair, to one block.

For example, if a dual word touches only blocks j and k, its two blocks w_j, w_k induce functionals $\beta_j, \beta_k \in V^*$ satisfying $\beta_j(u_j) + \beta_k(u_k) = 0$ for every outer word $u \in O$. The pair (β_j, β_k) is therefore an outer functional-dual word. The outer gate rules out a nonzero pair of such small support. Once both induced functionals vanish, $w_j, w_k \in I^\perp$; the inner gate then says that two nonzero blocks would already exceed the allowed weight.

The *functional dual* of O is

$$O^{\perp_{\mathrm{fun}}} = \left\{ (\beta_j)_{j\in J} \in (V^*)^J : \sum_{j\in J} \beta_j(u_j) = 0 \text{ for every } u \in O \right\}.$$

Its distance is the minimum number of nonzero component functionals in a nonzero tuple; its support therefore counts the outer blocks touched by the induced dual relation. This is intrinsic to V and does not choose field coordinates. We use the standard convention $d(\{0\}) = +\infty$, so the functional gate is vacuous when the functional dual is zero.

Theorem 6.1 (Complete repair-hypergraph transfer)**.** *Let $r \geq 0$. Suppose*

$$r + 1 < 2d(I^{\perp}) \quad \text{and} \quad d(O^{\perp_{\mathrm{fun}}}) \geq r + 2.$$

Then for every block $j \in J$ and inner coordinate $x \in K$,

$$\mathcal{H}^{(r)}_{(j,x)}(O \circ_e I) = \{\{j\} \times R : R \in \mathcal{H}^{(r)}_x(I)\}.$$

Consequently matching number, transversal number, inclusion-minimal repair clutter, edge counts, and the full incidence structure are preserved coordinate by coordinate.

The first inequality is the *inner block-confinement gate*: two nonzero inner-dual blocks would already be too heavy. The second is the *outer functional gate*: a relation touching at most $r + 1$ blocks must induce the zero functional on each one.

Proof. Take $w \in (O \circ_e I)^{\perp}$ with $\mathrm{wt}(w) \leq r + 1$ and write w_j for its blocks. Each w_j defines a functional $\beta_j \in V^*$ by pairing w_j with $e(v)$. Orthogonality to the concatenated code says $(\beta_j)_j \in O^{\perp_{\mathrm{fun}}}$. Its functional support has size at most $\mathrm{wt}(w) \leq r + 1$, below the outer functional-dual distance. Hence every β_j is zero, so each block w_j lies in $I^{\perp}$.

Every nonzero block now has weight at least $d(I^{\perp})$. If two blocks were nonzero, then

$$\mathrm{wt}(w) \geq 2d(I^{\perp}) > r + 1,$$

a contradiction. Thus a bounded dual word containing (j, x) is supported only in block j, where its helper support is an inner repair of x. This proves one inclusion. Conversely, extending any inner dual word by zero outside block j gives a dual word of every concatenation, proving the other inclusion. ☐

Corollary 6.2 (Exact mixed-locality transfer)**.** *Under the hypotheses of Theorem 6.1, let $0 \leq s \leq r$. For every (j, x), the concatenated coordinate has exact locality s if and only if the inner coordinate x has exact locality s.*

Proof. The inner gate at radius s follows from $s + 1 \leq r + 1 < 2d(I^{\perp})$, and the outer gate follows from $d(O^{\perp_{\mathrm{fun}}}) \geq r+2 \geq s+2$. Apply Theorem 6.1 at every radius at most s. Existence of an s-repair and nonexistence at every smaller radius are therefore both preserved. ☐

The inner gate does not require $d(I^{\perp}) = r+1$. For the full $q = 9$ seed, $d(C_9^{\perp}) = 3$ because of the axis triples, yet radius-three transfer is exact since $4 < 2 \cdot 3$.

6.1 Sharpness of the transfer gates

Proposition 6.3 (Uniform boundary of the transfer gates)**.** *Neither numerical hypothesis in Theorem 6.1 can be weakened uniformly, even among nonzero codes over $\mathbb{F}_3$.*

(a) At radius $r = 3$ there are an inner code and a full outer code with $r + 1 = 2d(I^{\perp})$ for which complete repair-hypergraph equality fails.

(b) At radius $r = 2$ there are an inner code satisfying $r + 1 < 2d(I^{\perp})$ and an outer code with functional-dual distance exactly $r+1$ for which complete repair-hypergraph equality fails.

Thus the strict inner inequality and the outer threshold $r + 2$ are best possible for the stated uniform theorem. This does not assert that either hypothesis is necessary for a fixed concatenation.

Proof. For both examples let $I \leq \mathbb{F}_3^2$ be the repetition code $I = \{(a, a) : a \in \mathbb{F}_3\}$. Its dual is generated by $(1, -1)$, so $d(I^{\perp}) = 2$.

For (a), use two outer symbols and take the full outer space. Its functional dual is zero, so it satisfies every outer gate. The concatenated dual word

$$(1, -1 \mid 1, -1)$$

has weight four and meets both blocks. With its first coordinate as target, the other three support coordinates form a radius-three repair which is not an embedded repair of one inner block. Here $4 = 2d(I^{\perp})$.

For (b), use three outer symbols and the single-parity-check outer code

$$O = \{(a, b, c) \in \mathbb{F}_3^3 : a + b + c = 0\}.$$

Its functional dual consists of constant triples of scalar functionals and has exact functional-dual distance three. In the concatenation, the word that is one in the first coordinate of each block and zero elsewhere is a dual word of weight three. Targeting its first coordinate gives a two-helper repair meeting all three blocks, so it is not an embedded inner repair. Meanwhile $3 < 2d(I^{\perp}) = 4$.

Both examples give strict failure of complete repair-hypergraph equality. The two-block dual-distance mechanism is the same elementary symmetric- difference obstruction used for distinct recovery sets, for example by Kurz–Yaakobi [8, Lemma 10(a)]; the point here is the exact uniform boundary of the complete-hypergraph transfer statement. □

7 The trace bridge and the field-nine lift

Let $L/\mathbb{F}$ be a finite separable extension and let $O \leq L^J$ be an L-linear outer code. Restricting scalars makes each outer symbol an $\mathbb{F}$-vector space. The functional-dual gate in Theorem 6.1 is then supplied by the trace pairing.

Theorem 7.1 (Trace-duality bridge)**.** *Every $\mathbb{F}$-linear functional $f : L \to \mathbb{F}$ has a unique representation*

$$f(x) = \mathrm{Tr}_{L/\mathbb{F}}(a_f x).$$

If $(f_j)_j$ belongs to the functional dual of the restricted-scalar code, then $(a_{f_j})_j \in O^{\perp}$. Moreover

$$f_j = 0 \quad \Longleftrightarrow \quad a_{f_j} = 0$$

at every coordinate. Hence

$$d((O|_{\mathbb{F}})^{\perp_{\mathrm{fun}}}) \geq d(O^{\perp}).$$

Proof. Separable trace is a nondegenerate bilinear pairing, giving the unique coefficient a_f and exact support preservation. If $u \in O$ and $c \in L$, functional orthogonality applied to $cu \in O$ gives

$$0 = \sum_j f_j(cu_j) = \mathrm{Tr}_{L/\mathbb{F}}\left(c\sum_j a_{f_j} u_j\right) \qquad \text{for every } c \in L.$$

Nondegeneracy forces $\sum_j a_{f_j} u_j = 0$. Thus the coefficient tuple lies in the ordinary L-dual of O, with the same support. $\square$

Now take $\mathbb{F} = \mathbb{F}_9$, $L = \mathbb{F}_{9^4} = \mathbb{F}_{6561}$, and use a linear equivalence between the four-dimensional $\mathbb{F}_9$-space L and the four-dimensional inner code C_9.

Corollary 7.2 (Finite extension-field lift)**.** *Let $O \le L^N$ be an $[N, K, D]_L$ code with $O \neq 0$ and $d(O^\perp) \ge 5$. Its concatenation with C_9 is an $\mathbb{F}_9$-linear code with*

$$[19N,\ 4K,\ \ge 8D]_9.$$

Its complete radius-three repair hypergraph is an embedded copy of the corresponding C_9 hypergraph in each block. Consequently, among its $19N$ coordinates, the exact blockwise profile is:

type	*multiplicity*	*locality*	(ν, τ)
cubic	$9N$	3	$(4, 7)$
finite axis	$9N$	2	$(6, 12)$
infinity axis	N	2	$(7, 13)$.

The guaranteed arbitrary-helper failure thresholds $\tau - 1$ are respectively 6, 11, *and* 12.

Proof. Restriction of scalars multiplies dimension by four and does not change symbol support. A nonzero outer word has at least D nonzero symbols, each of whose inner encodings has weight at least eight, so the concatenated distance is at least $8D$. Theorem 7.1 supplies functional-dual distance at least five, while $4 < 2d(C_9^\perp) = 6$. Apply Theorem 6.1, Corollary 6.2, and Table 2. $\square$

Corollary 7.3 (Completed finite extension-field lift)**.** *Let $O \le L^N$ be an $[N, K, D]_L$ code with $O \neq 0$ and $d(O^\perp) \ge 6$. Concatenation with $\overline{C}_9$ is an $\mathbb{F}_9$-linear code with*

$$[20N,\ 4K,\ \ge 9D]_9.$$

Its complete repair hypergraphs through radius four are embedded copies of the corresponding inner hypergraphs. Exactly $10N$ coordinates are cubic, with exact locality three and row $(4, 8)$ at radius four; exactly $10N$ are axis coordinates, with exact locality two and row $(7, 15)$. The guaranteed helper-failure thresholds are 7 *and* 14.

Proof. The parameter argument is the same, now using the $[20, 4, 9]_9$ inner seed. At radius four the inner gate is $5 < 2d(\overline{C}_9^\perp) = 6$, and the trace bridge requires ordinary outer dual distance $r+2 = 6$. Theorem 6.1 therefore gives equality of the complete bounded hypergraphs. Apply Theorem 5.2. As in Theorem 6.1, the conclusion concerns the radius-four repair hypergraph. $\square$

8 A fixed-alphabet asymptotic family

We require one deep existence theorem. Stichtenoth [13, Theorem 1.6(ii)] proves that for every square $Q = \ell^2$ there is an unbounded sequence of self-dual $[N_j, N_j/2, D_j]_Q$ codes with

$$\liminf_j \frac{D_j}{N_j} \geq \frac{1}{2} - \frac{1}{\ell - 1}.$$

For $Q = 6561 = 81^2$, the lower bound is $39/80$. Self-duality is especially useful here: the same D_j supplies both the primal distance needed for the concatenated distance and the dual distance needed for exact repair transfer.

Theorem 8.1 (Uniform asymptotic repair family)**.** *There is a sequence of* $\mathbb{F}_9$*-linear codes* $\mathcal{C}_j$ *with lengths tending to infinity such that, for all sufficiently large* j*,*

(a) $\mathcal{C}_j$ *has exact rate* $2/19$*;*

(b) its relative minimum distance is at least $1/5$*;*

(c) in every inner block, nine cubic coordinates have exact locality three, nine finite-axis coordinates exact locality two, and one infinity-axis coordinate exact locality two;

(d) on those three respective types, the complete radius-three repair rows are $(4, 7)$*,* $(6, 12)$*, and* $(7, 13)$*, and the guaranteed helper-failure thresholds are* 6*,* 11*, and* 12*.*

In particular $7\nu_x \leq 4\tau_x$ *at every coordinate. More sharply, for every fixed real* $c < 39/190$*, one has*

$$\frac{d(\mathcal{C}_j)}{\mathrm{length}(\mathcal{C}_j)} > c$$

for all sufficiently large j *(with the threshold allowed to depend on* c*).*

Proof. Take Stichtenoth's self-dual codes over $L = \mathbb{F}_{6561}$ and concatenate them with C_9 after restriction to $\mathbb{F}_9$. Two numerical consequences are needed: dual distance at least five for the transfer gate, and enough primal distance for the displayed relative-distance bound. Since $39/80 > 1/3$, eventually

$$N_j \leq 3D_j \quad \text{and} \quad N_j \geq 15.$$

The second inequality and the first imply $D_j \geq 5$, so self-duality gives $d(O_j^{\perp}) = D_j \geq 5$ and Corollary 7.2 applies. Since also $39/80 > 19/40$, eventually $19N_j \leq 40D_j$. The lifted length and dimension are $19N_j$ and $4(N_j/2) = 2N_j$, giving exact rate $2/19$. Its distance is at least $8D_j$, and

$$\frac{d(\mathcal{C}_j)}{19N_j} \geq \frac{8D_j}{19N_j} \geq \frac{1}{5}.$$

The exact localities and repair rows transfer blockwise.

For the sharper distance statement, fix $c < 39/190$. Then $19c/8 < 39/80$, so eventually $D_j/N_j > 19c/8$. Hence

$$\frac{d(\mathcal{C}_j)}{19N_j} \geq \frac{8D_j}{19N_j} > c.$$

□

Theorem 8.2 (Completed comparison family)**.** *There is a sequence of* $\mathbb{F}_9$*-linear codes* $\overline{\mathcal{C}}_j$ *of unbounded length and exact rate* $1/10$ *such that, eventually, the relative minimum distance is at least* $1/5$*. Each inner block has ten cubic coordinates of exact locality three and radius-four* *row* $(4,8)$*, and ten axis coordinates of exact locality two and row* $(7,15)$*. More sharply, for every fixed real* $c < 351/1600$*,*

$$\frac{d(\overline{\mathcal{C}}_j)}{\mathrm{length}(\overline{\mathcal{C}}_j)} > c$$

eventually, with the threshold allowed to depend on c*.*

Proof. Use the same self-dual outer family and concatenate with $\overline{C}_9$. Here the corresponding goals are dual distance at least six and the completed distance ratio. Eventually $N_j \geq 18$ and $N_j \leq 3D_j$, hence $D_j \geq 6$ and Corollary 7.3 applies. Length and dimension are $20N_j$ and $4(N_j/2) = 2N_j$, giving rate $1/10$. The distance is at least $9D_j$. Since $39/80 > 4/9$, eventually $4N_j < 9D_j$, which yields the clean $1/5$ bound.

For fixed $c < 351/1600 = 9(39/80)/20$, one has $20c/9 < 39/80$, so eventually

$$\frac{d(\overline{\mathcal{C}}_j)}{20N_j} \geq \frac{9D_j}{20N_j} > c.$$

The bounded radius-four hypergraph equality transfers the coordinate profiles. The conclusion remains the bounded radius-four equality supplied by Theorem 6.1. □

The tradeoff between the two constructions is summarized in Table 3. Completion slightly lowers the rate, raises the available asymptotic distance threshold, merges the axis subtypes, and transfers one additional repair radius.

	punctured seed	completed seed
inner parameters	$[19,4,8]_9$	$[20,4,9]_9$
exact asymptotic rate	$2/19$	$1/10$
eventual distance range	$c < 39/190$	$c < 351/1600$
transferred radius	3	4
block multiplicities	$9+9+1$	$10+10$
repair rows	$(4,7),(6,12),(7,13)$	$(4,8),(7,15)$

Table 3: Comparison of the two fixed-alphabet families. In the distance row, every fixed c in the displayed range is eventually a strict lower bound on relative distance.

9 Relation to prior work

The locality of codeword symbols and its parameter bounds originate in the modern LRC literature represented by [4]. Multiple repair alternatives and finite-geometric constructions are explicit in Pamies-Juarez–Hollmann–Oggier [11]; exact availability for large classes of generalized quadrangles is developed in Lavrauw–Van de Voorde [9]. In particular, [11, Definition 3] defines local repair tolerance by the same minimum hitting-set formula as our τ, while Wang–Zhang [14] organize several tolerance notions through regenerating sets. These works establish the tolerance interpretation; our results are the exact coordinatewise pair (ν,τ), its strict separation on every symbol of this family, and the transfer of the entire bounded repair family.

Gruica–Jany–Ravagnani [5] give a refined weight distribution indexed by prescribed subsets of dual-word supports and prove duality identities for it. This is close prior art for treating all bounded dual supports rather than a selected repair group. It does not state the matching/transversal computations here or a concatenation theorem that identifies the complete bounded support family.

For concatenation, Liu–Ma–Wu–Xing [10, Theorem 3.1] concatenate a locally repairable inner code with classical outer codes and derive locality and asymptotic parameter families. The recent construction of Jin–Fu [7] also uses concatenation to obtain binary LRCs with disjoint local repair groups and controlled weight distributions. These results establish locality and code parameters, not equality of every bounded repair support. Our transfer theorem strengthens the conclusion under an additional outer-dual gate: it identifies every bounded dual support and therefore transfers the complete repair hypergraph, not only the existence of an inner recovery set.

The geometry of the twisted cubic and its stabilizer has a substantial literature. Davydov–Marcugini–Pambianco [2] describe the axis in characteristic three and line orbits under the cubic stabilizer; Bartoli–Davydov–Marcugini–Pambianco [1] compute point–plane incidence data and multiple-covering-code parameters. Those incidence counts are adjacent to, but do not by themselves determine, matching and transversal numbers of the repair hypergraphs in Sections 4 and 5. The full projective cubic and its characteristic-three common axis supply the classical geometry; the new calculation concerns the exact full-minimal repair rows of their union.

The multiplicative pairing used in the cubic matching proof belongs to a classical design-theoretic neighborhood. Constant-quotient pairings of a finite-field half-set are studied as one-quotient starters; see, for example, Dinitz [3]. Here we combine that pattern with the shifted-inverse completion identity to compute the exact matching number of the code-derived repair hypergraph.

In summary, the contribution is the conjunction of an explicit all-symbol $\tau > \nu$ seed, exact complete-hypergraph transfer, and the resulting fixed-alphabet parameters.

10 Formal verification and provenance

The project was human-directed and agent-generated. The named author selected the problem, directed the research, set the validation standards, and takes responsibility for the final claims; AI systems generated substantial portions of the mathematics, code, formal proofs, and prose. AI is not listed as an author.

The theorem chain is formalized in Lean 4.32.0-rc1 with the matching pinned mathlib revision. The root module `RepairCodes` kernel-checks the finite geometry and repair rows, complete bounded repair-hypergraph transfer and its two sharpness examples, the support-preserving trace bridge and finite lifts, and every reduction from the asymptotic input to the displayed rates and distance bounds. In particular, `cubicRepair_matchingNumber` formalizes the colored-graph matching, `repairHypergraph_concatenatedCode_eq_embed` is the exact transfer equality, and `hasFunctionalDualDistanceAtLeast_restrictScalars` is the trace bridge. The formal `repairHypergraph` uses the exact dual-word support definition in Definition 2.1.

The independent script `notes/2026-07-13-projective-completion-verifier.py` implements its own finite fields and projective normalization. Deterministically and in exact arithmetic, it exhausts all projective plane forms at $q = 3, 9, 27$, checks the repair rows at $q = 3, 9$, and accepts only when the section spectra, parameters, circuit relation, and exact rows agree; two column mutations are required to fail the corresponding invariants. These executions reproduce evidence but are not consumed by the Lean proofs.

The formal chain contains no admitted declarations, unsafe definitions, native execution, code-generation trust, or externally generated proof certificates. Its finite computations use kernel reduction. Finite, algebraic, and analytic-reduction declarations depend only on `propext`, `Classical.choice`, and `Quot.sound`. The asymptotic theorems additionally consume one project axiom, `Imported.stichtenoth_selfDual_TVZ_6561`: the exact specialization of Stichtenoth's Theorem 1.6(ii) stated in Section 8.

The artifact entry point, pinned dependencies, reproduction commands, expected finite outputs, theorem map, and axiom reports are recorded in the paper-package README, proof ledger, and `lean/RepairCodes/TRUST.md`; the verifier above uses standard Python and completes without a specialized resource profile.

11 Conclusion

The complete bounded repair hypergraph separates two roles that ordinary locality compresses: a matching measures simultaneous disjoint repair, whereas a transversal measures resistance to arbitrary helper failures. The twisted-cubic–axis systems make this distinction explicit at every coordinate, and the concatenation theorem shows that the entire incidence structure—not merely one local repair—can persist in an asymptotically good family.

The two seeds also expose a concrete tradeoff. Puncturing gives the higher rate family and three coordinate types at radius three; projective completion raises the transferred radius, merges the axis types, and improves the available distance threshold at a small rate cost.

The sharp uniform gates may be unnecessary for fixed pairs. Open problems include bounding the matching–transversal gap and relating support transfer to subpacketized bandwidth and access.

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